

Nathan Gong

(609) 454-1896 | nathangong9@gmail.com | linkedin.com/in/nathangong9 | github.com/nathan-gong

EXPERIENCE

- Software Engineer II** | *Medtronic* | Boston, MA Jan 2022 - Present
- Develop production software for a surgical robotics platform in an embedded, mixed-OS (Linux, QNX, Windows) environment
 - Integrated a failsafe system that reduced error recovery time by 70% and enhanced reliability during clinical procedures, impacting hundreds of cross-functional stakeholders and accelerating FDA approval.
 - Refactored the core GUI framework onto the new failsafe architecture and built an alarm routing system to propagate messages across distributed compute nodes, enabling end-to-end testability and improving real-time error visibility.
 - Migrated platform software to a unified Linux OS, implementing core C++ library enhancements to harden system security through least-privilege service execution and application image authentication.
 - Built a full-stack Python application to aggregate web data and generate risk reports for 30+ off-the-shelf software products, improving report generation time by 80% and streamlining the user workflow.
 - Designed and validated manufacturing tooling for a production line transfer and strengthened build infrastructure with structured error handling and CI-integrated unit test suites.
- Teaching Assistant** | *Northeastern University* | Boston, MA Sep 2020 - Aug 2022
- Mentored 100+ students by providing technical guidance, debugging code, and explaining core algorithmic concepts.
- Software Engineer Co-op** | *iRhythm Technologies* | San Francisco, CA Jan 2021 - Jun 2021
- Automated firmware testing infrastructure for an ambulatory cardiac monitoring device, reducing test time by 50%
 - Designed APIs to interface with manufacturing software tools and configure embedded systems for production.

SKILLS

Languages: Python, C++, Java, JavaScript, SQL, C, Bash

Libraries/Frameworks: Flask, Spring Boot, Pytest, GoogleTest, PyTorch, NumPy, Pandas, Scikit-learn

Tools/Platforms: Git, Docker, GCP, Jira, Bamboo, Linux, QNX, Windows, MySQL, MongoDB, Neo4j, Jupyter

Practices: TDD, OOP, Agile, CI/CD, SDLC

EDUCATION

- Columbia University** | New York City, NY 2024 - 2026
- Master of Science in Computer Science. GPA: 3.97/4.0
- Coursework:* High-Performance ML, AI, Computer Vision, Computational Robotics, NLP, HCI, Databases, Networks
- Northeastern University** | Boston, MA 2019 - 2023
- Bachelor of Science in Bioengineering. Minor in Computer Science. GPA: 3.96/4.0
- Honors:* University Honors Program, President's Award, Tau Beta Pi, PEAK Research Award

PROJECTS

- Roofline-Aware LLM Inference Analysis** Jan 2026 - May 2026
- Optimized Llama-3.1-8B inference on an NVIDIA L4 GPU by implementing FlashAttention-2 and 4-bit quantization, achieving a 63% memory reduction at an 18% decode-throughput cost.
- Robotic Tree Pruning Simulator** Sep 2025 - Dec 2025
- Developed a motion-planning pipeline for autonomous robotic tree pruning using 3D point-cloud segmentation to localize cut points, surface reconstruction for collision modeling, and trajectory planning for the robotic arm.
- Movie Data Visualization Application** Nov 2024 - Dec 2024
- Built a RESTful API and ETL pipeline to scrape, transform, and serve thousands of movie records, persisting data across relational/document/graph stores and powering interactive visualizations of popularity trends over time.